

CPO / NPO vs VCSEL

Which architecture actually solves the networking bottleneck

Technical Primer | Optical Interconnect | Beginner-oriented, academically sourced

1 EXECUTIVE SUMMARY

ELI5 — The question, restated

Imagine a huge office where every desk has to talk to every other desk. Today the desks are wired together with copper cables that plug into the wall at the front of the room. The office keeps getting bigger, the cables keep getting hotter, and the plugs at the front of the room are now the thing holding everything back. Two ideas compete. Idea one: move the plug from the wall right onto the desk, so the copper run is millimetres instead of metres. That is co-packaged optics. Idea two: change what kind of light bulb you use inside the plug - swap the expensive, precision-built bulb for a cheap one you can print thousands of at a time on a wafer. That is the VCSEL. The critical thing to understand before anything else is that these are answers to two different questions, and they are not mutually exclusive.

Core question: Where the optics sits, and what makes the light	Pluggable optics today: ~15-30 pJ/bit (energy per bit)
CPO demonstrated: ~5-10 pJ/bit; sub-1 pJ/bit claimed in research	Best VCSEL CPO paper: 4.5 pJ/bit, 8 x 106 Gb/s, JLT 44(3) 2026
Switch ASIC cadence: Capacity doubles roughly every 2 years	Copper reach at 224G: ~1 m passive DAC; tens of cm at 448G
Multimode reach at 200G/lane: ~30-50 m over OM4 fibre	Single-mode CPO reach: 500 m to 2 km (DR-class links)
Volume CPO shipping: Broadcom TH6-Davisson (Oct 2025)	NVIDIA CPO timing: Quantum-X early 2026; Spectrum-X H2 2026
Standards position: OIF CPO PRD specifies single-mode	Report date: 12 August 2026

The framing 'CPO/NPO versus VCSEL' contains a category error that is worth dissolving before any analysis, because almost every confused argument in this debate traces back to it. Co-Packaged Optics (CPO) and Near-Packaged Optics (NPO) are packaging architectures: they describe how far the optical engine sits from the switch or accelerator chip. A Vertical-Cavity Surface-Emitting Laser (VCSEL) is a light source: it describes what physically generates the photons. These are orthogonal axes. You can build a CPO module whose light source is a VCSEL, and in 2026 several vendors have done exactly that.

The genuine technical contest, once the axes are separated, is between two full-stack system choices. Option A is single-mode silicon photonics co-packaged optics: an external continuous-wave laser feeds light into a silicon photonic integrated circuit sitting on the switch package, where micro-ring or Mach-Zehnder modulators encode the data onto single-mode fibre. This is what NVIDIA, Broadcom and TSMC have industrialised. Option B is multimode VCSEL-based near-package or co-packaged optics: a two-dimensional array of directly modulated VCSELs sits next to the host chip and drives multimode fibre over short distances. This is what Lumentum, Coherent and Broadcom's VCSEL group are proposing for scale-up networks inside a rack.

The evidence base supports a split outcome rather than a winner. The published energy and reach data show single-mode silicon photonics CPO winning decisively in scale-out networks, where links run hundreds of metres between racks and rows, because multimode fibre physically cannot carry 200 Gb/s per lane beyond roughly 30 to 50 metres. VCSEL arrays win on manufacturing cost, thermal tolerance and supply chain independence in scale-up networks, where links are under about 10 metres and the workload is dominated by moving tensor data between accelerators inside one rack. The most likely 2027-2030 steady state is both, in different layers of the same machine.

This primer builds the argument from the physics upward. Section 2 explains what an optical link is made of, with no assumed background. Section 3 quantifies why the bottleneck exists now and not five years ago. Sections 4 and 5 take the two axes - packaging and light source - separately. Section 6 puts them head to head against published measurements. Sections 7 to 9 cover the ecosystem, the risks, and what evidence to watch next. A full bibliography of the cited literature is Appendix D.

2 PHYSICS FOUNDATIONS - WHAT AN OPTICAL LINK IS MADE OF

ELI5 — An optical link, end to end

Sending data over fibre is like flashing a torch down a very long, very narrow pipe at someone at the other end. You need five things. Something that decides when to flash (the driver electronics). A light source (the torch bulb). A way of turning the light on and off fast (either flicking the bulb itself, or leaving the bulb on and putting a shutter in front of it). The pipe (the fibre). And someone at the far end watching for the flashes and writing down the message (the photodetector and receiver electronics). Every argument in this report is about which of those five things you choose and where you physically put them.

Formally, a datacentre optical link consists of a serializer/deserializer (SerDes) that converts wide, slow parallel data inside the chip into a narrow, fast serial stream; a driver circuit; a light source; a modulator (which may be the light source itself); an optical fibre; a photodiode (PD); a transimpedance amplifier; and a receiver SerDes. Modern links additionally carry Forward Error Correction (FEC), which adds redundant bits so the receiver can repair errors, and in most current designs a Digital Signal Processor (DSP) that equalises the distorted electrical waveform.

Two modulation schemes matter here. Non-Return-to-Zero (NRZ) uses two light levels, off and on, carrying one bit per symbol. Four-Level Pulse Amplitude Modulation (PAM4) uses four light levels, carrying two bits per symbol, so it doubles the data rate at the same symbol rate (measured in Gigabaud, or GBd). The cost is signal-to-noise ratio: with four levels squeezed into the same optical power range, the gaps between levels are one third as large, which costs roughly 9.5 dB of noise margin and forces heavier FEC and DSP. This trade-off is the hinge of the whole VCSEL argument, as Section 5 shows.

ELI5 — Single-mode versus multimode fibre

Multimode fibre is a wide pipe. Light can bounce down it along many different paths, some straight down the middle, some zig-zagging off the walls. The zig-zagging light takes a longer route and arrives late, so a sharp flash at one end arrives smeared at the other. Single-mode fibre is a pipe so narrow - about a fifth the width - that only one path exists, so the flash arrives sharp. The narrow pipe is harder and more expensive to aim light into, which is the entire reason anyone still uses the wide one.

Multimode fibre (MMF) has a core diameter of about 50 micrometres and supports many spatial propagation modes. Because each mode has a different group velocity, a pulse spreads in time as it propagates - this is modal dispersion, and it is the dominant reach limit for MMF. Graded-index profiles in OM3, OM4 and OM5 fibre partially compensate by making the refractive index highest at the core centre, so off-axis light travels faster and catches up. The residual capability is quantified as Effective Modal Bandwidth (EMB), expressed in MHz-km. OM4 is specified at 4,700 MHz-km at 850 nm against 2,000 MHz-km for OM3.

Single-mode fibre (SMF) has a core of roughly 9 micrometres and supports a single spatial mode, eliminating modal dispersion entirely and leaving only chromatic dispersion, which is far weaker over datacentre distances. The penalty is coupling: aligning a light source to a 9 micrometre core requires sub-micron mechanical precision and, in most manufacturing flows, active alignment - powering the device up and moving it under feedback until the light lands. Multimode coupling into a 50 micrometre core tolerates several micrometres of misalignment and can be done passively, which is roughly an order of magnitude

cheaper in assembly cost. This one manufacturing fact drives most of the cost argument for VCSELs.

ELI5 — Energy per bit, the universal scorecard

If you want to compare two ways of sending data fairly, do not compare watts, because a faster link naturally uses more power. Compare how much energy it takes to move one single bit. That number is measured in picojoules per bit - a picojoule is one trillionth of a joule. It is the industry's miles-per-gallon figure, and every claim in this report reduces to it.

Energy per bit (pJ/bit) is total electrical power consumed divided by the data rate. It is the correct comparison metric because it normalises across generations and architectures. Published figures should be read carefully: some vendors quote only the optical engine, others include the SerDes, the DSP and the laser, and the difference between those accounting boundaries is often larger than the difference between the technologies being compared. Where this report cites a figure, the accounting boundary is stated or flagged in Appendix A.

Component	What it does	ELI5 equivalent	Where the debate touches it
SerDes	Converts wide parallel on-chip data to a fast serial stream and back	The narrow doorway everything must file through	Its power and its physical width at the chip edge are the bottleneck
Driver	Supplies the current or voltage swing that encodes bits onto light	The finger on the light switch	Directly modulated VCSELs need a simpler driver than external modulators
Light source	Generates photons - VCSEL, EML or continuous-wave laser	The torch bulb	This is the VCSEL axis of the debate
Modulator	Encodes data onto light if the source is not directly modulated	A shutter in front of the bulb	Micro-ring modulators are tiny but thermally fragile
Fibre	Carries the light - multimode or single-mode	The pipe	Multimode is cheap to couple but short-reach; single-mode is the reverse
Photodiode	Converts arriving light back into electrical current	The eye watching for flashes	Array photodiodes pair naturally with VCSEL arrays
DSP	Digitally equalises the distorted waveform	A spell-checker for smeared signals	CPO and LPO both aim to delete it; deleting it saves most of the power
FEC	Adds redundancy so errors can be repaired	Saying it twice so it survives a bad line	KP4 FEC is the threshold most published link demos are measured against

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THE BOTTLENECK, QUANTIFIED - WHY NOW AND NOT FIVE YEARS AGO

ELI5 — The shoreline problem

Picture a chip as an island. All the data that leaves the island has to depart from the beach around its edge - the shoreline. The island's interior keeps getting more productive every generation, because transistors keep shrinking. But the beach does not get longer, because the chip is already about as big as a reticle allows. So you have a factory doubling its output every two years and a port that stays the same size. Eventually the port, not the factory, sets how much you can ship. That is where AI networking is now.

The core measurement comes from Minkenberg and colleagues in IET Optoelectronics (2021), who documented that Ethernet switch application-specific integrated circuit (ASIC) capacity has doubled roughly every two years, achieved by alternately increasing the number of SerDes lanes and the signalling rate per lane. Both levers are now near exhaustion. Lane count is limited by shoreline: the physical perimeter of the die available for SerDes macros. Signalling rate is limited by channel loss: pushing a copper lane faster makes every via, bump, trace and connector bleed more signal.

The consequence is a power inversion that the same paper documents. The fraction of switch ASIC power consumed by SerDes has risen with each generation to the point where the power used for optical links is about to exceed that of the switch core itself. Their cost observation is equally striking: because a large fraction of optical module cost is packaging and assembly, which does not scale with semiconductor process technology the way ASIC capacity does, the optics required to populate a datacentre switch now cost more than the switch.

Reach compounds the problem. At 224 Gb/s per lane, the reach of passive Direct Attach Copper (DAC) cable is approximately one metre; at a nominal 448 Gb/s it is expected to fall to tens of centimetres. This is why NVLink scale-up domains are confined to a single rack, and why NVIDIA's route to higher aggregate bandwidth in NVLink 6.0 is reported to be bidirectional SerDes - sharing one physical copper channel between a 224 Gb/s transmit and a 224 Gb/s receive path - rather than a true doubling of unidirectional lane rate. That is a clever workaround, not a new headroom.

Constraint	Physical cause	Current numbers	Why it forces optics
Shoreline density	SerDes macros occupy die perimeter; reticle limit caps perimeter	Order of 0.4 Tb/s per mm unidirectional in current disclosures	Optical I/O escapes vertically through the package instead of laterally
Channel loss	Skin effect, dielectric loss, reflections at every discontinuity	Roughly 20 dB lost at 212 Gb/s before the connector; over 40 dB budgets for long-reach SerDes	Loss grows exponentially with reach in copper, near-flat in fibre
SerDes power share	Equalisation complexity rises superlinearly with channel loss	Approaching or exceeding switch core power	Shortening the electrical path to millimetres removes most of the equalisation burden
Copper reach	Signal integrity collapse at high baud	About 1 m at 224G; tens of cm expected at 448G	Rack-scale scale-up domains cannot grow without optics
Faceplate area	Front-panel cage count is bounded by chassis geometry	A 102.4T switch cannot physically host enough pluggable cages at acceptable power density	Optics must move off the faceplate onto or near the package

The scale of the prize is best shown by a deployment arithmetic that circulates widely in the industry: in a cluster of roughly 128,000 accelerators, moving to a co-packaged architecture is reported to cut the number of pluggable transceivers from close to 500,000 to around 128,000. NVIDIA's own framing puts the electrical loss reduction at roughly 22 dB down to 4 dB and per-port power at roughly 30 W down to 9 W. Both figures are vendor-sourced and should be treated as directional rather than independently verified; they are flagged as such in Appendix A.

4 **AXIS ONE - WHERE THE OPTICS SITS (PLUGGABLE, LPO, NPO, CPO)**

ELI5 — Four places to put the plug

Start with a lamp on your desk plugged into a socket on the far wall, with a long extension lead full of noise filters. That is a pluggable transceiver. Now throw away the noise filters and hope the signal survives - that is Linear Pluggable Optics. Now move the socket from the far wall to the edge of your desk - that is Near-Packaged Optics. Now build the socket into the lamp base itself - that is Co-Packaged Optics. Each step shortens the noisy copper run and saves power, and each step makes it harder to swap the lamp when it breaks.

The Optical Internetworking Forum (OIF) draws the boundary precisely. Co-packaging means attaching the optical device to the same first-level substrate as the host ASIC. A socketed arrangement of separately packaged devices sharing a common board-level substrate is near-package optics. Pluggable optics places removable modules in cages on the switch front panel. The distinction is the physical integration boundary, and it matters commercially because it determines whether a failed optical component can be replaced in the field by hand.

Architecture	Where the optics sits	Electrical path	Energy per bit	Serviceability	Status mid-2026
Pluggable (DSP)	Front-panel cage	15-30 cm of PCB trace	~15-30 pJ/bit	Hot-swappable in minutes, multi-vendor	Default; still the volume architecture
LPO / LRO	Front-panel cage, DSP removed	Same trace, ASIC SerDes drives the optic directly	Captures much of the CPO saving	Unchanged - still hot-swappable	The pragmatic 2026-27 choice for most networks
NPO	Separate substrate or socket beside the ASIC	Short copper on a shared board	Between LPO and CPO	Optical engine remains socketable and detachable	Intermediate step; favoured for VCSEL scale-up proposals
CPO	Same first-level package as the ASIC	Millimetres	~5-10 pJ/bit demonstrated	Failed engine can condemn the whole ASIC package	Shipping: Broadcom TH6-Davisson, NVIDIA Quantum-X
In-package optical I/O	Optical chiplet inside the ASIC package, 3D integrated	Sub-millimetre	Research targets below 1 pJ/bit	Not field serviceable at all	Early; Ayar Labs and academic demonstrations

The energy claims are consistent across independent sources, which is unusual and gives some confidence. Broadcom's published figures put pluggable modules at 15 to 20 pJ/bit against 5 to 10 pJ/bit for CPO. The Minkenberg paper computed 25 to 30 pJ/bit for then-current 400G transceivers, with 7 nm DSP chips expected to bring 400G and 800G modules below 20 pJ/bit, and set a CPO target of 10 to 15 pJ/bit for the first generation with better to follow. Broadcom's TH6-Davisson, using sixteen 6.4 Tb/s silicon optical engines built on TSMC's Compact Universal Photonic Engine (COUPE) process, is quoted at roughly 70% lower optical interconnect power.

Two mechanisms produce that saving, and it is worth separating them because only one is unique to co-packaging. The first is deleting the DSP: once the electrical channel is short enough, the receiver no longer needs heavy digital equalisation. Linear Pluggable Optics (LPO) captures this saving without any packaging change, which is precisely why it is the commercially dominant near-term answer. The second is deleting the long-reach SerDes drive itself, which only co-packaging achieves. Analysts framing the economics have put five-year interconnect cost for a 100,000-accelerator build at roughly US\$1.44 billion for pluggables, US\$480 million for LPO and US\$315 million for CPO - vendor-adjacent modelling, directional only, but it explains why LPO is treated as the cash cow and CPO as the endpoint.

The reliability objection is the serious one, and it has a published research history. The Stanford work on external versus integrated light sources (Journal of Lightwave Technology, vol. 39 no. 7, 2021) framed it explicitly: because servicing or replacing highly integrated devices is expensive, and because those devices may fail at a higher rate due to elevated temperatures, switches may need to be designed to tolerate failed optical links, with under-provisioning or internal switching to backup ports. That paper adopted a conservative tolerable annual failure rate of 1% from optical components. In CPO, a failed optical engine can strand an entire premium ASIC package - the 'blast radius' problem - which is why every serious commercial implementation has made the laser a field-replaceable unit. Broadcom uses pluggable, replaceable laser modules precisely because light sources have the highest failure rate.

Counter-evidence exists on the other side of that argument. Broadcom's case for CPO is not only power but link stability: integrating the optical engines removes many sources of manufacturing and test variability inherent to pluggable transceivers, which the company says improves link flap performance, citing a study on the prior-generation TH5-Bailly engine. In AI training, where a single flapping link can idle a large fraction of a synchronous job, stability may prove worth more than swappability. This is an empirical question that field data over 2026-27 will settle, and it is the single most important thing to watch (Section 9).

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AXIS TWO - WHAT MAKES THE LIGHT (VCSEL, EML, CW PLUS MODULATOR)

ELI5 — Two ways to make a flashing light

You can flick the bulb on and off - simple, cheap, but bulbs do not like being flicked billions of times a second and they get sloppy when hot. Or you can leave the bulb on permanently and put a very fast shutter in front of it - the bulb stays happy, the shutter can be made microscopically small on a silicon chip, but now you have two components instead of one, and the shutter is fussy about temperature. The first is a directly modulated VCSEL. The second is a continuous-wave laser feeding a silicon photonic modulator.

A Vertical-Cavity Surface-Emitting Laser (VCSEL) is a semiconductor laser diode that emits perpendicular to the wafer surface. The active region - typically strained indium gallium arsenide quantum wells - is sandwiched between two Distributed Bragg Reflector (DBR) mirrors, each built from dozens of alternating quarter-wavelength layers of aluminium gallium arsenide and gallium arsenide (GaAs). Because the optical cavity is shorter than a single wavelength, the device naturally operates in one longitudinal mode. An oxide-confined aperture, formed by wet oxidation of a high-aluminium layer, simultaneously confines current and the optical mode, which is why threshold currents below one milliampere are achievable.

Three engineering consequences follow directly from that structure, and they are the entire commercial case for VCSELs. First, emission is perpendicular to the surface, so devices can be fabricated and tested at wafer level and arranged into two-dimensional arrays - unlike edge-emitting lasers, which must be cleaved into bars before they emit at all. Second, the device is directly modulated: the drive current itself carries the data, so no separate modulator, no external laser, and a much simpler driver. Third, the emitted beam has a large divergence and couples naturally into a 50 micrometre multimode core, permitting passive alignment.

The alternative architecture separates the functions. A continuous-wave (CW) laser, usually indium phosphide (InP) based, supplies constant optical power. That light enters a silicon Photonic Integrated Circuit (PIC), where either a Mach-Zehnder Modulator (MZM) or a Micro-Ring Modulator (MRM) encodes the data. An MZM splits light into two arms, shifts the phase in one, and recombines them so the interference produces amplitude modulation - robust and broadband, but physically large, typically millimetres. An MRM is a ring waveguide a few tens of micrometres in circumference that resonates at one wavelength; shifting the resonance by carrier injection or depletion switches transmission on and off. MRMs are two to three orders of magnitude smaller and lower power, which is why NVIDIA's Quantum-X photonic engines use 200 Gb/s MRMs on TSMC COUPE.

ELI5 — Why micro-rings are fragile

A micro-ring works like a wine glass that sings at exactly one note. Change the temperature of the glass a little and the note shifts. Now imagine gluing that wine glass to the lid of a running laptop and asking it to hold pitch. That is what co-packaging a micro-ring next to a 1,000-watt AI chip asks of it.

Quantitatively: silicon's thermo-optic coefficient produces a resonance shift of roughly 0.1 nm per degree Celsius. Since a micro-ring's usable resonance window is narrow - practical operating ranges for ring modulators are often cited as under one degree Celsius, against thirty degrees or more for Mach-Zehnder and electro-absorption devices - every ring needs an integrated heater and a closed-loop controller. Padmaraju and Bergman's review in Nanophotonics (2014) remains the canonical treatment; it reports tuning powers as low as roughly 20 mW per free spectral range with adiabatic ring designs and integrated heaters, and microsecond tuning speeds. That heater power is a standing tax on the energy-per-bit budget that VCSELs simply do not pay, and Intel engineers have publicly flagged that thermal fluctuation from the co-packaged GPU or switch die is a first-order functional risk, not a second-order performance one.

Against that, VCSELs carry their own physics tax: modal dispersion in multimode fibre. The literature is precise. Work published in Photonics (2025) on launch condition optimisation states that at 200 Gb/s per lane, VCSEL to multimode fibre transmission reaches are reduced to 30 to 50 metres, and that effective modal bandwidth above the OM4 threshold is required to exceed 50 metres at all. The mitigation paths in

the literature are launch-condition engineering, high-data-rate OM4 variants with better modal bandwidth near 850 to 920 nm, and single-mode or multi-aperture single-mode VCSELs that narrow the spectral width.

Property	Directly modulated VCSEL	CW laser plus silicon MRM/MZM	Which wins and why
Components in the link	One active device per lane	Laser, coupler, modulator, heater, controller	VCSEL - fewer failure modes, simpler bill of materials
Fibre coupling	Passive alignment into 50 um core	Active alignment into 9 um core or edge coupler	VCSEL - roughly an order of magnitude cheaper assembly
Wafer economics	GaAs, wafer-level test, 2D arrays native	Silicon-on-insulator in CMOS fabs plus separate InP laser	Split - VCSEL cheaper per device; SiPh scales function count better
Thermal tolerance	1060 nm platforms specified above 150 C operation	Micro-rings shift ~0.1 nm per degree C; need active stabilisation	VCSEL decisively, and this matters most under a hot ASIC
Reach at 200G per lane	About 30-50 m over OM4	500 m to 2 km on single-mode DR-class links	Silicon photonics decisively
Wavelength multiplexing	Limited; SWDM over a narrow window	Dense WDM natural - many wavelengths down one fibre	Silicon photonics - fibre count scales far better
Laser reliability exposure	Laser is the data device; no separate source to fail	External laser is the highest-failure component, hence made field-replaceable	VCSEL structurally simpler; SiPh mitigates by making lasers FRUs
Standards support	OIF CPO product requirements specify single-mode, effectively excluding VCSELs	Squarely inside the OIF CPO definition	Silicon photonics - a real commercial disadvantage for VCSEL CPO

The wavelength shift in VCSEL roadmaps deserves attention because it is the industry quietly conceding part of the reach argument while strengthening the thermal one. Historic datacom VCSELs operate at 850 nm on GaAs. Both Lumentum and Coherent are moving scale-up products to 1060 nm. Longer wavelength improves speed capability and high-temperature performance, and because GaAs is transparent at 1060 nm it permits backside emission - light exits through the substrate - which in turn permits flip-chip assembly directly onto a package. Lumentum's OFC 2026 demonstration integrates 1060 nm VCSELs and photodiodes in a two-dimensional monolithic array with backside lenses, in a fan-out wafer-level package, explicitly targeting 'slow and wide' scale-up protocols such as UCle and PCle, and supporting channel sparing for reliability.

6 HEAD TO HEAD - WHAT THE MEASUREMENTS ACTUALLY SAY

ELI5 — Reading a link demonstration paper

When a paper says '106 Gb/s PAM4 over 50 m of OM4 below the KP4 FEC threshold', it is saying: we sent symbols at 106 billion per second, two bits each, down fifty metres of a specific grade of wide fibre, and the number of errors was low enough that the standard error-correction scheme could clean them all up. Every number in that sentence is load-bearing. Change the fibre grade, the distance, or the FEC assumed, and the claim changes completely. This is why comparing headline speeds between press releases is almost meaningless.

The most directly relevant published result for this comparison is a Journal of Lightwave Technology paper (vol. 44, no. 3, 2026) describing a high-density, energy-efficient 850 nm VCSEL-based eight-channel CPO transceiver at 106 Gb/s per channel for AI and machine learning applications. Its measurements: bit error rate below the KP4 FEC threshold across a 100 m OM4 loopback with all eight channels running simultaneously, total power of 3.8 W, and link energy of 4.5 pJ/bit, with an equaliser integrated into the VCSEL driver to meet the CEI-112G-XSR-PAM4 specification. That 4.5 pJ/bit sits at or below the bottom of the 5 to 10 pJ/bit band commonly quoted for silicon photonics CPO - which is the single strongest quantitative datapoint in the VCSEL case.

Two caveats keep that result from being decisive. First, the accounting boundary is a transceiver, not a full switch port including the host SerDes, so it is not like-for-like against system-level vendor figures. Second, 100 m OM4 at 106 Gb/s per channel is close to the multimode ceiling; the modal bandwidth literature indicates that at 200 Gb/s per lane the same fibre supports only 30 to 50 m. A technology that wins on energy at 50 m and cannot reach 300 m at all is not a general-purpose answer - it is a scale-up answer.

On the VCSEL device side, the Broadcom programme is the most thoroughly published. Their extended paper in Photonics (vol. 13 no. 1, article 90, January 2026) reports 850 nm oxide VCSELs with -3 dB bandwidth above 35 GHz and relative intensity noise below -152 dB/Hz, targeting 106 GBd PAM4 for 200 Gb/s per lane, with no uncorrectable errors over 50 m of OM4. The development history it records is instructive about pace: 200 Gb/s multimode links with uncorrected bit error rate below 1e-7 at ECOC in September 2024, improved to roughly 1e-8 at OFC in March 2025, and a nine-hour stability run over 50 m OM4 with at most five correctable frame errors. Notably, the paper's title itself proposes VCSELs for near-package optics scale-up - the device community has already conceded the scale-out layer.

Related device records worth knowing: 850 nm single-lane 200 Gb/s PAM4 over 100 m OM4 using noise cancellation, partial response signalling and a simplified Volterra nonlinear equaliser (IEEE Photonics Technology Letters, 2021); 168 Gb/s PAM4 over 50 m using 28 GHz multimode VCSELs (Lavrencik et al., OFC 2020); 128 Gb/s NRZ direct modulation of a 1060 nm single-mode VCSEL over 2 km of standard single-mode fibre (OECC 2023); and a 24.9 GHz bandwidth VCSEL supporting 170 Gb/s with advanced modulation formats (Photonics Research, 2024). The single-mode 1060 nm results matter disproportionately: they show a directly modulated VCSEL reaching kilometre distances on single-mode fibre, which if manufacturable at scale would erase the reach objection.

On the silicon photonics side, the comparable device-level datapoints are a five-channel 200 Gb/s micro-ring modulator silicon chip using two-segment z-shape junctions (Nature Communications, 2024), Broadcom's 32-channel full-duplex optical engine at 100 Gb/s per channel for 3.2 Tb/s aggregate with 5 to 10 pJ/bit and only 2 to 3 dB of ASIC-to-engine electrical insertion loss, and research transmitters reaching 0.78 pJ/bit at 64 GBd using a photonic-crystal slow-light modulator with a co-designed current-mode driver (arXiv:2506.04820). The last of these is the honest counterweight to the VCSEL energy argument: the theoretical floor for integrated silicon photonics is well below anything a directly modulated laser can reach, because a modulator does not have to pay for generating photons.

Dimension	Single-mode SiPh CPO	Multimode VCSEL NPO/CPO	Verdict
Demonstrated energy per bit	5-10 pJ/bit system level; sub-1 pJ/bit in research devices	4.5 pJ/bit at transceiver level (JLT 2026)	VCSEL leads today at comparable integration; SiPh has the lower theoretical floor
Reach	500 m to 2 km DR-class	30-50 m at 200G/lane over OM4	SiPh - not close
Thermal robustness next to a hot die	Micro-rings need heaters and closed-loop control	1060 nm platforms rated beyond 150 C	VCSEL - a genuine structural advantage
Assembly cost	Active alignment, sub-micron tolerance	Passive alignment, wafer-level test, 2D arrays	VCSEL
Fibre count per Tb/s	DWDM collapses many lanes onto one fibre	One fibre per lane; wide-and-slow means very high fibre counts	SiPh - and this is underrated as a serviceability issue
Supply chain	InP lasers, SOI wafers, advanced packaging (COUPE/CoWoS)	GaAs wafers reusing the 3D-sensing manufacturing base	VCSEL - independent and higher-volume base
Standards and ecosystem	OIF CPO PRD written around single-mode	Outside the current CPO standard; aligning instead to UCle/PCle scale-up	SiPh
Best-fit network layer	Scale-out: leaf-spine, rack-to-rack, row-to-row	Scale-up: accelerator-to-accelerator inside a rack	Complementary, not substitutable

7

SUPPLY CHAIN AND ECOSYSTEM MAPPING

ELI5 — Two different factories

The two approaches do not just make different products - they are built in different kinds of factory, from different raw materials, by different companies. Silicon photonics borrows the world's most advanced chip factories and adds a separate specialist laser bought in from elsewhere. VCSELs are made on their own dedicated material in factories that already stamp out billions of units a year for phone face-unlock sensors. That difference in industrial base is a strategic argument, not just a technical one.

Input	Silicon photonics CPO	VCSEL NPO/CPO	Concentration risk
Substrate	Silicon-on-insulator wafers in CMOS foundries	Gallium arsenide wafers	SOI supply is narrow; GaAs epitaxy is specialised but multi-sourced
Light source	External continuous-wave InP lasers, made field-replaceable	The VCSEL is the source - no separate laser	InP and indium are geopolitically exposed; GaAs less so
Advanced packaging	TSMC COUPE, CoWoS-class integration; 3D stacking of EIC on PIC	Fan-out wafer-level packaging using standard semiconductor flows	COUPE/CoWoS capacity is the tightest single chokepoint in the chain
Fibre and connectors	Single-mode fibre arrays, precision connectors (Corning, SENKO)	Multimode fibre; passive-alignment connectors	Fibre attach and connector supply is an underappreciated bottleneck for both
Switch silicon	Broadcom Tomahawk 6, NVIDIA Spectrum-6, Marvell Teralynx	Host ASIC agnostic; targets UCle and PCIe scale-up interfaces	Very high concentration in switch ASIC supply either way
Test and burn-in	Known-good-die and laser burn-in are the critical yield gates	Wafer-level test is native to the VCSEL process	Known-good-die discipline is the main CPO yield risk

The strategic asymmetry is worth stating plainly. Silicon photonics CPO concentrates dependency on the single most contested resource in the industry - leading-edge advanced packaging capacity - and adds an indium phosphide laser supply chain on top. The VCSEL route leans on a gallium arsenide manufacturing base that already ships in extreme volume for consumer three-dimensional sensing; Lumentum states its 1060 nm platform is built on a base that has shipped more than ten billion emitters. Vendors describe this explicitly as offering an independent supply chain alternative to silicon photonics. For a buyer worried about single-point dependency on one foundry's packaging roadmap, that is a real consideration and not merely marketing.

Ecosystem participants worth tracking, without any investment view attached: on the silicon photonics CPO side, Broadcom (Tomahawk 6 Davission, sixteen 6.4 Tb/s engines), NVIDIA (Quantum-X and Spectrum-X Photonics), TSMC (COUPE), Coherent (silicon photonics collaboration and InP lasers), Lumentum (lasers for Spectrum-X), Corning and SENKO (fibre and connectivity), and Marvell. On the VCSEL side, Lumentum and Coherent both have 1060 nm backside-emitting array programmes, and Broadcom's VCSEL group is the most published source of 200 Gb/s per lane device data. Ayar Labs represents the third path - in-package optical input/output as a chiplet - which sits beyond CPO on the integration spectrum and argues that current OIF-form-factor CPO is really near-package optics with poor electronic-photonic co-integration.

8 CATALYSTS, RISKS AND OPEN RESEARCH QUESTIONS

Catalyst	Timing	Impact	Probability
NVIDIA Spectrum-X Photonics volume ramp	H2 2026	First large-scale Ethernet CPO deployment; field reliability data becomes observable	High
TH6-Davission field flap and failure statistics from hyperscale operators	H2 2026 - H1 2027	Settles the blast-radius objection empirically; the single highest-information event	Medium
IEEE 802.3dj completion and 200G-per-lane multimode objectives	2026-2027	Determines whether multimode gets a standardised 200G/lane path or is left to proprietary scale-up	Medium
OIF revisiting multimode within the CPO framework	2027+	Would remove the largest structural disadvantage facing VCSEL CPO	Low

Catalyst	Timing	Impact	Probability
UCle or PCIe optical scale-up specification maturing	2026-2028	Creates the standards home VCSEL NPO currently lacks; Lumentum is explicitly targeting this	Medium
Single-mode 1060 nm VCSEL manufacturability at scale	2027+	Would erase the reach objection entirely and reopen scale-out for VCSELs	Low
LPO capturing the 1.6T generation outright	2026-2027	Delays CPO adoption by a full generation; the historical base rate for CPO slipping	High
Sub-1 pJ/bit integrated modulator moving from research to product	2028+	Would end the energy argument decisively in favour of integrated silicon photonics	Low

Risk one - the blast radius and the field-replaceable-unit compromise. In a pluggable architecture a failed module degrades one port. In CPO, a failed optical engine can render an expensive switch package unusable, and troubleshooting is materially harder. Every credible commercial design has responded by making the laser - the highest-failure component - a field-replaceable unit, which is a partial fix that leaves modulators, photodiodes and fibre attach non-serviceable. The Stanford light-source study framed this as requiring switches designed to tolerate failed links, through under-provisioning or internal switchover to backup ports, and assumed a 1% tolerable annual optical failure rate. If real-world CPO rates land above that, the architecture does not fail outright, but the system design overhead materially erodes the power advantage.

Risk two - micro-ring thermal control under AI workloads. This is the specific failure mode most likely to surprise. Silicon's roughly 0.1 nm per degree Celsius resonance shift means a co-packaged ring modulator sitting beside a switch ASIC or accelerator must be actively stabilised while the neighbouring die swings through large power transients as workloads start and stop. Intel engineers have publicly stated that package thermal fluctuation can affect functionality, not merely performance. Recent work has gone as far as proposing that the software scheduler give the photonic bias controller advance warning of inference load spikes tens of milliseconds ahead. A control loop that depends on the workload scheduler is a fragile place for a network to live, and this is precisely where the VCSEL's 150 C-plus tolerance is a genuine structural advantage rather than a marketing claim.

Risk three - the multimode reach wall is physics, not engineering effort. At 200 Gb/s per lane the measured VCSEL to OM4 reach is 30 to 50 metres, and exceeding 50 metres requires effective modal bandwidth above the OM4 specification. Launch-condition engineering, high-data-rate OM4 variants and single-mode multi-aperture VCSELs all push at this, but none of them changes the underlying fact that many spatial modes with different group velocities smear a pulse. Any thesis that depends on VCSELs winning scale-out networks, where links run hundreds of metres between racks and rows, is arguing against measured modal bandwidth data.

Risk four - standards exclusion is a commercial risk independent of the physics. The OIF co-packaged optics product requirements documents are written around single-mode operation and compact module geometries, which effectively excludes VCSEL technology from the CPO standard as currently drafted. Standardisation is what gave pluggables their multi-vendor interoperability and pricing pressure. A VCSEL scale-up ecosystem that lives outside OIF and instead attaches itself to UCle and PCIe is betting on a different standards home forming in time. That may work, but it is a bet on ecosystem formation rather than on device physics.

Risk five - LPO eats the transition window. Linear Pluggable Optics removes the DSP while keeping the familiar hot-swappable, multi-vendor module, capturing much of the power saving with none of the serviceability cost. Industry commentary describes it as plug-and-play, low-risk and immediately deliverable, and as the most realistic choice for the 2026-27 window, with the 1.6T switches shipping from major original design manufacturers built on traditional architectures. Broadcom itself hedges by offering a standard, purely electrical Tomahawk 6 alongside Davisson. CPO has been forecast as imminent and then deferred repeatedly since roughly 2018; the base rate favours another deferral at least in the scale-out layer.

Risk six - fibre count and connector burden in a wide-and-slow world. The VCSEL scale-up architecture wins by being wide: many parallel lanes at moderate rates, one fibre per lane. Single-mode silicon photonics can collapse many lanes onto one fibre through wavelength division multiplexing. At rack scale, the wide-and-slow approach implies very large fibre counts, and fibre attach, connector loss and physical cable management are already flagged in the literature as underappreciated CPO bottlenecks. This is not a physics objection but an operations one, and operations objections are what actually kill datacentre architectures.

Open research questions, stated as the literature leaves them. Can adiabatic micro-ring designs with in-resonator heaters hold sub-nanometre resonance under realistic AI power transients without a scheduler hint? Can single-mode 1060 nm VCSELs, which have already demonstrated 128 Gb/s NRZ over 2 km of single-mode fibre in the laboratory, be manufactured with the yield and uniformity that array products require? Does the measured field failure rate of co-packaged optical engines land above or below the 1% annual threshold assumed in the reliability literature? And what is the correct accounting boundary for pJ/bit comparisons - until the industry standardises one, cross-vendor energy claims are not comparable.

9 WHAT MOVES IT NEXT - EVIDENCE AND EARNINGS SIGNAL MAP

ELI5 — Why this section exists

A debate like this does not get settled by argument; it gets settled by a small number of observable events. Rather than re-reading the whole literature every quarter, work out in advance which single number each event will produce, and watch only that. This table lists the events, when they land, the one number to extract, and what it would mean.

Event	When	Days away	The one metric	Why it matters
Coherent (COHR) FQ4 FY26 results	12 Aug 2026 (today, after close)	0	Datacom commentary on VCSEL array sampling versus silicon photonics CPO content per port	Coherent supplies both architectures. Which one it flags as ramping is the cleanest read available on where design wins are actually landing.
Lumentum (LITE) FQ4 FY26 results	11 Aug 2026 (reported)	-1	Whether the 1060 nm VCSEL scale-up platform is described as customer-qualified or still evaluation-stage	Lumentum is the most committed public advocate of VCSEL scale-up. Qualification language is the difference between a demonstration and a programme.
NVIDIA (NVDA) Q2 FY27 results	26 Aug 2026 (confirmed)	~14	Any timing revision to Spectrum-X Photonics availability in H2 2026	A slip would be the strongest single signal that CPO is deferring another generation and that LPO owns the 1.6T window.
Broadcom (AVGO) FQ3 FY26 results	Early Sep 2026 (est.)	~25	TH6-Davisson unit commentary and any disclosed link-flap or failure statistics	Broadcom is the only vendor with two prior CPO generations in the field. Its reliability data is the most credible evidence on the blast-radius question.
OFC 2027 technical programme	March 2027	~215	Whether 200G/lane multimode papers extend reach beyond 50 m over OM4	The reach wall is the binding constraint on VCSELs. Movement here changes the addressable layer; no movement confirms scale-up confinement.
IEEE 802.3dj and successor project status	Rolling through 2027	-	Presence or absence of a standardised 200G-per-lane multimode objective	Standards inclusion determines whether VCSEL scale-up gets multi-vendor interoperability or stays proprietary.

Prioritisation. Broadcom's FQ3 print in early September is the one to read most carefully, because Broadcom is the only participant with two generations of co-packaged engines already in customer hands, and the blast-radius reliability question is the only objection to CPO that cannot be resolved by physics arguments alone - it needs field hours. NVIDIA on 26 August is second, purely for the Spectrum-X Photonics timing line. Coherent tonight is third but arrives first, and is useful mainly as a read on which architecture is generating design wins rather than demonstrations. For a beginner building intuition, the highest-yield single act is to

read the Photonics 13(1) 90 paper and the JLT 44(3) 955 paper side by side: between them they contain the entire VCSEL case, with measurements rather than claims.

A VERIFICATION LOG

Claim	Status	Note
Pluggable 15-30 pJ/bit; CPO 5-10 pJ/bit	EST.	Consistent across Broadcom-sourced figures, the Minkenberg paper and multiple secondary reviews. Accounting boundaries differ between sources; treat as a band, not a point.
VCSEL CPO transceiver at 4.5 pJ/bit, 3.8 W, 8 x 106 Gb/s	CLEAN	Journal of Lightwave Technology vol. 44 no. 3 p.955 (2026) abstract, Optica Publishing Group.
200G VCSEL: >35 GHz bandwidth, RIN below -152 dB/Hz, 50 m OM4	CLEAN	Photonics vol. 13 no. 1 article 90 (Jan 2026), MDPI, open access.
200G/lane VCSEL-MMF reach of 30-50 m	CLEAN	Photonics vol. 12 no. 7 article 654 (2025), stated explicitly in the introduction.
Silicon resonance shift ~0.1 nm per degree C	CLEAN	Cross-confirmed in two independent sources including Amkor engineering commentary.
Micro-ring usable temperature window under 1 degree C	FLAGGED	Sourced from a patent specification comparing modulator types. Directionally consistent with the ring-tuning literature but not from a peer-reviewed measurement.
Switch ASIC capacity doubling roughly every 2 years	CLEAN	Minkenberg et al., IET Optoelectronics (2021), doi:10.1049/ote2.12020.
Copper reach ~1 m at 224G, tens of cm at 448G	EST.	arXiv:2510.15893 and corroborating industry commentary. Forward projection for 448G.
NVIDIA 3.5x power efficiency, 22 dB to 4 dB loss, 30 W to 9 W per port	EST.	Vendor-sourced. Note NVIDIA materials quote both 3.5x (Mar 2025) and 5x (2026 product pages) for power efficiency; presented as a range, not resolved.
Broadcom TH6-Davisson ~70% optical interconnect power reduction	EST.	Vendor-sourced; sixteen 6.4 Tb/s engines and 64 Condor 3 nm SerDes cores confirmed via Broadcom release and independent trade coverage.
Five-year interconnect cost US\$1.44bn / US\$480m / US\$315m	FLAGGED	Third-party analyst modelling for a 100,000-accelerator build. Directional illustration only; assumptions not disclosed.
128,000-GPU cluster: ~500,000 pluggables to ~128,000	EST.	Widely repeated industry arithmetic; original derivation not traced to a primary source.
OIF CPO PRD effectively excludes VCSELs	EST.	Reported consistently in secondary technical coverage; the OIF documents themselves were not fetched directly.
Lumentum OFC 2026 1060 nm scale-up demonstration specifications	CLEAN	Lumentum investor relations press release, 17 March 2026.
Earnings dates: LITE 11 Aug, COHR 12 Aug, NVDA 26 Aug 2026	CLEAN	LITE and COHR confirmed via trade coverage of the current reporting week; NVDA confirmed.
Broadcom FQ3 FY26 date	EST.	Estimated from prior-year cadence; not confirmed by company IR at time of writing.
Bibliography author lists	FLAGGED	Several entries in Appendix D give title, journal, volume and year only, because search extracts did not confirm the full author list. Verify authorship before citing formally.
Company-level financial data	N/A	This is a technology primer. No market caps, valuations, price targets or recommendations are presented, consistent with house convention for thematic work.

B ACRONYM GLOSSARY

Acronym	Full name and one-line meaning
ASIC	Application-Specific Integrated Circuit - a chip built for one job, here the switch chip
BER	Bit Error Rate - the fraction of bits received wrongly
CoWoS	Chip-on-Wafer-on-Substrate - TSMC advanced packaging technology

Acronym	Full name and one-line meaning
COUPE	Compact Universal Photonic Engine - TSMC's silicon photonics integration process
CPO	Co-Packaged Optics - optics on the same first-level package as the host chip
CW	Continuous-Wave - a laser that stays on constantly rather than flashing
DAC	Direct Attach Copper - a passive copper cable assembly
DBR	Distributed Bragg Reflector - a stack of thin layers acting as a mirror
DSP	Digital Signal Processor - digitally cleans up a distorted signal
DWDM	Dense Wavelength Division Multiplexing - many wavelengths down one fibre
EMB	Effective Modal Bandwidth - how much bandwidth a multimode fibre really supports
EML	Externally Modulated Laser - laser and modulator in one package
FEC	Forward Error Correction - redundant bits that let the receiver repair errors
FRU	Field-Replaceable Unit - a part a technician can swap on site
GaAs	Gallium Arsenide - the semiconductor VCSELs are grown on
GBd	Gigabaud - billions of symbols per second
InP	Indium Phosphide - the semiconductor most datacom lasers are grown on
KP4 FEC	The Reed-Solomon error correction scheme most high-speed Ethernet links assume
LPO	Linear Pluggable Optics - a pluggable module with the DSP removed
MMF	Multimode Fibre - wide-core fibre, cheap to couple, short reach
MRM	Micro-Ring Modulator - a tiny resonant ring that switches light on and off
MZM	Mach-Zehnder Modulator - an interference-based modulator, large but robust
NPO	Near-Packaged Optics - optics beside the chip but on a separate substrate or socket
NRZ	Non-Return-to-Zero - two-level signalling, one bit per symbol
OIF	Optical Internetworking Forum - the standards body defining CPO interfaces
OM3/OM4/OM5	Grades of laser-optimised multimode fibre, in ascending modal bandwidth
PAM4	Four-Level Pulse Amplitude Modulation - two bits per symbol
PD	Photodiode - converts light back into electrical current
PIC	Photonic Integrated Circuit - an optical chip
pJ/bit	Picojoules per bit - energy consumed to move one bit; the universal scorecard
RIN	Relative Intensity Noise - laser output power noise, limits achievable BER
SerDes	Serializer/Deserializer - converts parallel on-chip data to a fast serial stream
SiPh	Silicon Photonics - optical circuits built in silicon using CMOS processes
SMF	Single-Mode Fibre - narrow-core fibre, one propagation path, long reach
SOI	Silicon-on-Insulator - the wafer type used for silicon photonic circuits
SWDM	Short Wavelength Division Multiplexing - a few closely spaced wavelengths on MMF
TDECQ	Transmitter Dispersion Eye Closure Quaternary - a transmitter quality metric
UCIe	Universal Chiplet Interconnect Express - a chiplet-to-chiplet interface standard
VCSEL	Vertical-Cavity Surface-Emitting Laser - a laser emitting through the wafer surface
WPE	Wall-Plug Efficiency - electrical power in versus optical power out
XPU	Generic term for a compute accelerator - GPU, TPU or custom silicon

C DISCLAIMER

This document is for internal research purposes only and does not constitute investment advice, an offer, or a solicitation to buy or sell any security. It contains no price targets and no buy, sell or hold recommendations. Companies are named only as participants in the technology ecosystem under discussion. Technical figures are drawn from published academic literature and vendor disclosures as cited; vendor figures are labelled as such and should be treated as directional. All data should be independently verified before use in any investment decision.

D BIBLIOGRAPHY - PRIMARY LITERATURE CITED

Entries marked with a dagger give title, journal and year only because the full author list was not confirmed from available extracts. Verify authorship before formal citation.

Ref	Citation	What it establishes
1	Minkenberg, C., et al. Co-packaged datacenter optics: opportunities and challenges. IET Optoelectronics, 2021. doi:10.1049/ote2.12020	The foundational systems paper: ASIC capacity doubling, SerDes power share, cost inversion, and CPO energy targets
2	External vs. Integrated Light Sources for Intra-Data Center Co-Packaged Optical Interfaces. Journal of Lightwave Technology 39(7), Apr 2021 (Kahn group, Stanford). (dagger)	The reliability and link-budget framework; the 1% tolerable annual optical failure assumption
3	A High-Density Energy-Efficient 850-nm VCSEL-Based 106-Gb/s x 8-Channel CPO Transceiver for AI/ML Applications. Journal of Lightwave Technology 44(3), 955, 2026. (dagger)	The strongest VCSEL datapoint: 4.5 pJ/bit, 3.8 W, KP4-compliant over 100 m OM4
4	200G VCSEL Development and Proposal of Using VCSELs for Near-Package-Optics Scale-Up Application. Photonics 13(1), 90, 2026. doi:10.3390/photonics13010090	Broadcom device data at 106 GBd PAM4; and the explicit proposal of VCSELs for NPO scale-up
5	Wang, J., Murty, M., Giovane, L., et al. 200Gb/s PAM4 oxide VCSEL development progress at Broadcom. Proc. SPIE VCSELs XXIX, 1138402, 2025	The conference original underlying reference 4
6	Murty, M.V.R., et al. Toward 200G per Lane VCSEL-based Multimode Links. OFC 2024	The roadmap statement of intent for 200G/lane multimode
7	Modal Bandwidth Enhancement Through Launch Condition Optimization for High Data Rate VCSEL Transmission Over Multimode Fibers. Photonics 12(7), 654, 2025. (dagger)	The reach wall: 30-50 m at 200G/lane, and the launch-condition mitigation path
8	Link Bandwidth and Transmission Capability of Single-Mode Multi-Aperture VCSELs at 100 G/Lane and 200 G/Lane over Multimode Fibers. Photonics 12(2), 147, 2025. (dagger)	Single-mode multi-aperture VCSELs as the reach mitigation
9	Wavelength Dependence of Modal Bandwidth of Multimode Fibers for High Data Rate Transmission and Its Implications. Photonics 11(7), 667, 2024. (dagger)	High-data-rate OM4 variants and the case for shifting VCSEL centre wavelength
10	850-nm VCSEL-Based Single-Lane 200-Gbps PAM-4 Transmission for Datacenter Intra-Connections. IEEE Photonics Technology Letters 33(19), 1042, 2021. (dagger)	200 Gb/s over 100 m OM4 using Volterra nonlinear equalisation
11	Lavrencik, J., Varughese, S., Ledentsov, N., Chorchos, L., Ledentsov, N.N., Ralph, S.E. 168Gbps PAM-4 Multimode Fiber Transmission through 50m using 28GHz 850nm Multimode VCSELs. OFC 2020, paper W1D.3	A widely cited multimode capacity benchmark
12	Shimizu, S., et al. Record High-baud-rate 128-Gbit/s NRZ-OOK Direct Modulation of 1060-nm Single-mode VCSEL for Transmission over 2-km Standard SMF. OECC 2023	The key counterexample: a VCSEL reaching kilometre distances on single-mode fibre
13	Wu, W.-L., Cheng, C.-H., Matsumoto, A., Akahane, K., Yang, Y.-C., Su, B., Wu, C.-H., Feng, M., Lin, G.-R. 24.9-GHz-bandwidth VCSEL enables 170-Gbit/s OFDM, GFDM and DMT transmissions. Photonics Research 12, 377-390, 2024	VCSEL bandwidth records with advanced modulation formats
14	Padmaraju, K., Bergman, K. Resolving the thermal challenges for silicon microring resonator devices. Nanophotonics 3(4), 269-281, 2014	The canonical treatment of micro-ring thermal drift and heater-based stabilisation
15	Ophir, N., Mineo, C., Mountain, D., Bergman, K. Silicon photonic microring links for high-bandwidth-density, low-power chip I/O. IEEE Micro 33(1), 54-67, 2013	Early framing of micro-ring links as the route to high bandwidth density

Ref	Citation	What it establishes
16	Yuan, Y., Peng, Y., Sorin, W.V., Cheung, S., Huang, Z., Liang, D., Fiorentino, M., Beausoleil, R.G. A 5 x 200 Gbps microring modulator silicon chip empowered by two-segment z-shape junctions. Nature Communications 15, 918, 2024	State-of-the-art micro-ring modulator performance
17	Shekhar, S., Bogaerts, W., Chrostowski, L., Bowers, J.E., Hochberg, M., Soref, R., Shastri, B.J. Roadmapping the next generation of silicon photonics. Nature Communications 15, 751, 2024	The community roadmap for silicon photonics scaling
18	Tan, M., et al. Co-packaged optics (CPO): status, challenges and solutions. Frontiers of Optoelectronics, 2023	A comprehensive CPO review from the device and packaging side
19	Heterogeneous Integration Technology Drives the Evolution of Co-Packaged Optics. Micromachines 16(9), 1037, 2025. doi:10.3390/mi16091037 (dagger)	Coupling loss, packaging strategies, and the Broadcom 32-channel 3.2 Tb/s engine data
20	Progress in Research on Co-Packaged Optics. Review article, 2024 (PMC11509299). (dagger)	Standards history and system-level simulation results for all-to-all communication
21	High-efficiency compact optical transmitter with a total bit energy of 0.78 pJ/bit including silicon slow-light modulator and open-collector current-mode driver. arXiv:2506.04820, 2025. (dagger)	The low-energy frontier for integrated silicon photonic transmitters
22	Thermal management in co-packaged optics: from device assembly to system operation. Review article, 2025. (dagger)	Systematic treatment of CPO thermal coupling and reliability risk
23	Accelerating Frontier MoE Training with 3D Integrated Optics. arXiv:2510.15893. (dagger)	Shoreline and electrical reach limits quantified in an AI-training context